

Performance Testing

Date	02 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Developer Tools (Chrome), Python time module
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Prediction API (/predict)
Test Environment	Local System (Flask Development Server)
Test Date	01 july 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single prediction request	1	30	Response generated within 2 seconds
2	Multiple prediction requests	10	60	All requests processed successfully
3	Invalid input testing	5	30	Validation error displayed without crashing
4	Continuous prediction requests	20	120	Stable performance with no failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	1.2 sec	Pass	Within limit
3	Throughput (Req/sec)	≥ 15 req/sec	22 req/sec	Pass	Good performance

4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	32%	Pass	Acceptable
6	Memory Utilization	< 80%	41%	Pass	Stable usage

Step 4: Observations & Analysis

Key Findings

- Prediction API responded quickly under normal load.
- No failures occurred during testing.
- Model prediction remained accurate under repeated requests.

Bottlenecks Identified

- Slight increase in response time when multiple users send requests simultaneously.
- Flask development server is not suitable for production-scale traffic.

Optimization Steps Taken

- Optimized data preprocessing.
- Cached trained ML model in memory.
- Reduced unnecessary computations before prediction.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

AUTOMATED DECISIONING PIPELINE

Credit Card Approval Prediction Platform

Leveraging Multi-Algorithmic Machine Learning Classifiers to Streamline Credit Risk Appraisals

[Access Portal Form →](#)

Project Architecture & Scope

Financial systems require scalable alternatives to manual checks. This platform converts granular applicant demographics and historical credit balances into high-fidelity evaluations using a serialized predictive engine. The back-end handles data parsing, cleaning feature offsets, and addresses class imbalances dynamically before computing eligibility indices.

Implemented Engine Algorithms:

Model Engine Status

Preprocessing: Categorical Label Encoding mapped cleanly to Python integer vectors.

Feature Adjustments: Native negative-offset handling applied across `DAYS_BIRTH` & `DAYS_EMPLOYED`.

Platform Integration: Serialization via Joblib/Pickle connected to real-time Flask triggers.



Evaluation Entry Form

Fields pass directly to serialized scikit-learn preprocessing pipelines.

[← Return Dashboard](#)

1. CLEANED DEMOGRAPHIC ENCODINGS

Gender Label

Choose Label Option

Car Ownership Status

Does applicant own vehicle?

Realty Assets Status

Does applicant own real estate?

Income Stream Configuration

Select clean income mapping

Education Level Class

Select clean degree mapping

Civil Family Condition

Select current family mapping

Housing Operational Matrix

Select clear asset mapping

Occupation Profile Group

Select Encoded Job Cluster

2. QUANTITATIVE FINANCIAL BALANCE METRICS

Total Gross Income (\$)

Count of Dependent Children

Age Vector Duration (Days)



APPLICATION APPROVED

Risk exposure score sits within standard operational thresholds.

EVALUATION EXECUTION SUMMARY

Binary Risk Status Mapping (TARGET)

Standard Client (Class 0)

Assessed Model Archetype

Random Forest Classifier

Evaluation Execution Frame

02-07-2026 12:06:47

Process New Profile

[Go Back to System Dashboard](#)